

Team Attribute

2026-10-03

Load Data

```
team_attributes <- read.csv("Team_Attributes.csv")  
dim(team_attributes)
```

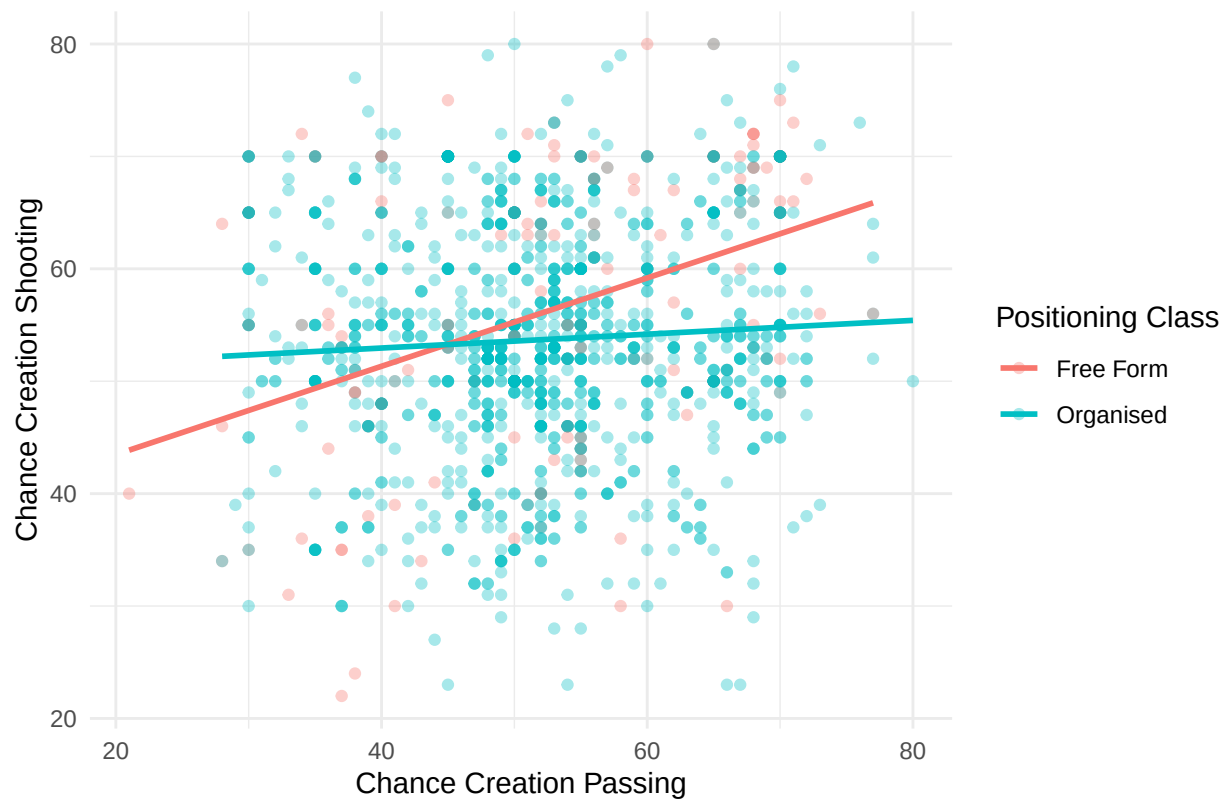
```
## [1] 1458  25
```

Interaction Plot

```
library(ggplot2)  
  
ggplot(team_attributes,  
  aes(x = chanceCreationPassing,  
      y = chanceCreationShooting,  
      color = chanceCreationPositioningClass)) +  
  geom_point(alpha = 0.35) +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(  
    x = "Chance Creation Passing",  
    y = "Chance Creation Shooting",  
    color = "Positioning Class",  
    title = "Passing vs. Shooting by Positioning Style"  
  ) +  
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Passing vs. Shooting by Positioning Style



Data Preparation

```
soccer_model_data <- team_attributes[, c(
  "chanceCreationShooting",
  "buildUpPlaySpeed",
  "buildUpPlayDribblingClass",
  "buildUpPlayPassing",
  "buildUpPlayPositioningClass",
  "chanceCreationPassing",
  "chanceCreationCrossing",
  "chanceCreationPositioningClass",
  "defencePressure",
  "defenceAggression",
  "defenceTeamWidth",
  "defenceDefenderLineClass"
)]

# Convert categorical variables to factors
soccer_model_data$buildUpPlayDribblingClass <-
  factor(soccer_model_data$buildUpPlayDribblingClass)

soccer_model_data$buildUpPlayPositioningClass <-
  factor(soccer_model_data$buildUpPlayPositioningClass)

soccer_model_data$chanceCreationPositioningClass <-
  factor(soccer_model_data$chanceCreationPositioningClass)
```

```

soccer_model_data$defenceDefenderLineClass <-
  factor(soccer_model_data$defenceDefenderLineClass)

# Check model dataset
dim(soccer_model_data)

## [1] 1458    12

colSums(is.na(soccer_model_data))

##           chanceCreationShooting           buildUpPlaySpeed
##                0                0
##    buildUpPlayDribblingClass    buildUpPlayPassing
##                0                0
##    buildUpPlayPositioningClass    chanceCreationPassing
##                0                0
##    chanceCreationCrossing chanceCreationPositioningClass
##                0                0
##           defencePressure           defenceAggression
##                0                0
##    defenceTeamWidth    defenceDefenderLineClass
##                0                0

```

Multiple Linear Regression Model

```

soccer_model <- lm(
  chanceCreationShooting ~
    buildUpPlaySpeed +
    buildUpPlayDribblingClass +
    buildUpPlayPassing +
    buildUpPlayPositioningClass +
    chanceCreationPassing *
    chanceCreationPositioningClass +
    chanceCreationCrossing +
    defencePressure +
    defenceAggression +
    defenceTeamWidth +
    defenceDefenderLineClass,
  data = soccer_model_data
)

summary(soccer_model)

##
## Call:
## lm(formula = chanceCreationShooting ~ buildUpPlaySpeed + buildUpPlayDribblingClass +
##    buildUpPlayPassing + buildUpPlayPositioningClass + chanceCreationPassing *
##    chanceCreationPositioningClass + chanceCreationCrossing +
##    defencePressure + defenceAggression + defenceTeamWidth +
##    defenceDefenderLineClass, data = soccer_model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -34.701  -5.761   0.052   6.733  28.169

```

```

##
## Coefficients:
##
## (Intercept) 29.19512
## buildUpPlaySpeed 0.08882
## buildUpPlayDribblingClassLots -3.25873
## buildUpPlayDribblingClassNormal -3.07549
## buildUpPlayPassing -0.08058
## buildUpPlayPositioningClassOrganised -2.17009
## chanceCreationPassing 0.35053
## chanceCreationPositioningClassOrganised 16.94787
## chanceCreationCrossing -0.05395
## defencePressure 0.11329
## defenceAggression 0.05146
## defenceTeamWidth 0.06075
## defenceDefenderLineClassOffside Trap 5.59984
## chanceCreationPassing:chanceCreationPositioningClassOrganised -0.31738
##
## Std. Error
## (Intercept) 4.03799
## buildUpPlaySpeed 0.02527
## buildUpPlayDribblingClassLots 2.16505
## buildUpPlayDribblingClassNormal 0.57023
## buildUpPlayPassing 0.02750
## buildUpPlayPositioningClassOrganised 1.28316
## chanceCreationPassing 0.06542
## chanceCreationPositioningClassOrganised 3.80952
## chanceCreationCrossing 0.02439
## defencePressure 0.03186
## defenceAggression 0.02954
## defenceTeamWidth 0.03121
## defenceDefenderLineClassOffside Trap 1.09012
## chanceCreationPassing:chanceCreationPositioningClassOrganised 0.06989
##
## t value Pr(>|t|)
## (Intercept) 7.230 7.80e-13
## buildUpPlaySpeed 3.514 0.000455
## buildUpPlayDribblingClassLots -1.505 0.132504
## buildUpPlayDribblingClassNormal -5.393 8.07e-08
## buildUpPlayPassing -2.930 0.003445
## buildUpPlayPositioningClassOrganised -1.691 0.091012
## chanceCreationPassing 5.358 9.79e-08
## chanceCreationPositioningClassOrganised 4.449 9.30e-06
## chanceCreationCrossing -2.212 0.027154
## defencePressure 3.555 0.000390
## defenceAggression 1.742 0.081714
## defenceTeamWidth 1.947 0.051757
## defenceDefenderLineClassOffside Trap 5.137 3.17e-07
## chanceCreationPassing:chanceCreationPositioningClassOrganised -4.541 6.05e-06
##
## (Intercept) ***
## buildUpPlaySpeed ***
## buildUpPlayDribblingClassLots ***
## buildUpPlayDribblingClassNormal ***
## buildUpPlayPassing **
## buildUpPlayPositioningClassOrganised .

```

```
## chanceCreationPassing ***
## chanceCreationPositioningClassOrganised ***
## chanceCreationCrossing *
## defencePressure ***
## defenceAggression .
## defenceTeamWidth .
## defenceDefenderLineClassOffside Trap ***
## chanceCreationPassing:chanceCreationPositioningClassOrganised ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.724 on 1444 degrees of freedom
## Multiple R-squared:  0.1214, Adjusted R-squared:  0.1135
## F-statistic: 15.35 on 13 and 1444 DF, p-value: < 2.2e-16
```

Key Model Results

```
# Full coefficient table
coef(summary(soccer_model))
```

	Estimate
## (Intercept)	29.19511715
## buildUpPlaySpeed	0.08881588
## buildUpPlayDribblingClassLots	-3.25872532
## buildUpPlayDribblingClassNormal	-3.07549404
## buildUpPlayPassing	-0.08058191
## buildUpPlayPositioningClassOrganised	-2.17008977
## chanceCreationPassing	0.35052975
## chanceCreationPositioningClassOrganised	16.94787300
## chanceCreationCrossing	-0.05394599
## defencePressure	0.11328539
## defenceAggression	0.05145825
## defenceTeamWidth	0.06074968
## defenceDefenderLineClassOffside Trap	5.59983635
## chanceCreationPassing:chanceCreationPositioningClassOrganised	-0.31738089
## Std. Error	
## (Intercept)	4.03799457
## buildUpPlaySpeed	0.02527325
## buildUpPlayDribblingClassLots	2.16504687
## buildUpPlayDribblingClassNormal	0.57023061
## buildUpPlayPassing	0.02750398
## buildUpPlayPositioningClassOrganised	1.28315749
## chanceCreationPassing	0.06542307
## chanceCreationPositioningClassOrganised	3.80952156
## chanceCreationCrossing	0.02439293
## defencePressure	0.03186309
## defenceAggression	0.02953907
## defenceTeamWidth	0.03120545
## defenceDefenderLineClassOffside Trap	1.09011860
## chanceCreationPassing:chanceCreationPositioningClassOrganised	0.06988517
## t value	
## (Intercept)	7.230103
## buildUpPlaySpeed	3.514225
## buildUpPlayDribblingClassLots	-1.505152

```
## buildUpPlayDribblingClassNormal -5.393421
## buildUpPlayPassing -2.929827
## buildUpPlayPositioningClassOrganised -1.691211
## chanceCreationPassing 5.357892
## chanceCreationPositioningClassOrganised 4.448819
## chanceCreationCrossing -2.211542
## defencePressure 3.555379
## defenceAggression 1.742040
## defenceTeamWidth 1.946765
## defenceDefenderLineClassOffside Trap 5.136906
## chanceCreationPassing:chanceCreationPositioningClassOrganised -4.541462
## Pr(>|t|)
## (Intercept) 7.796590e-13
## buildUpPlaySpeed 4.546573e-04
## buildUpPlayDribblingClassLots 1.325036e-01
## buildUpPlayDribblingClassNormal 8.066976e-08
## buildUpPlayPassing 3.444776e-03
## buildUpPlayPositioningClassOrganised 9.101234e-02
## chanceCreationPassing 9.787550e-08
## chanceCreationPositioningClassOrganised 9.296516e-06
## chanceCreationCrossing 2.715446e-02
## defencePressure 3.896129e-04
## defenceAggression 8.171435e-02
## defenceTeamWidth 5.175660e-02
## defenceDefenderLineClassOffside Trap 3.174467e-07
## chanceCreationPassing:chanceCreationPositioningClassOrganised 6.051340e-06

# Model fit statistics
summary(soccer_model)$r.squared

## [1] 0.1213915

summary(soccer_model)$adj.r.squared

## [1] 0.1134816

# Sample size used in the model
nobs(soccer_model)

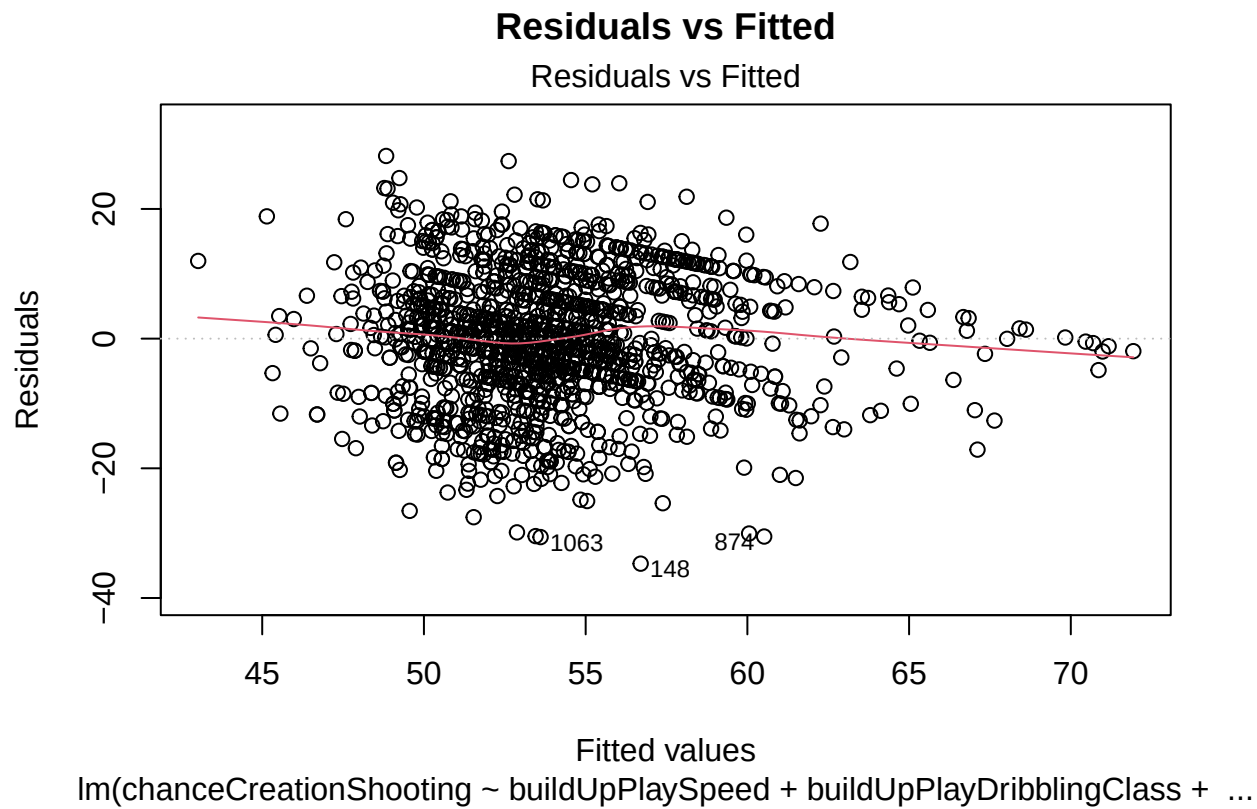
## [1] 1458
```

Model Diagnostics

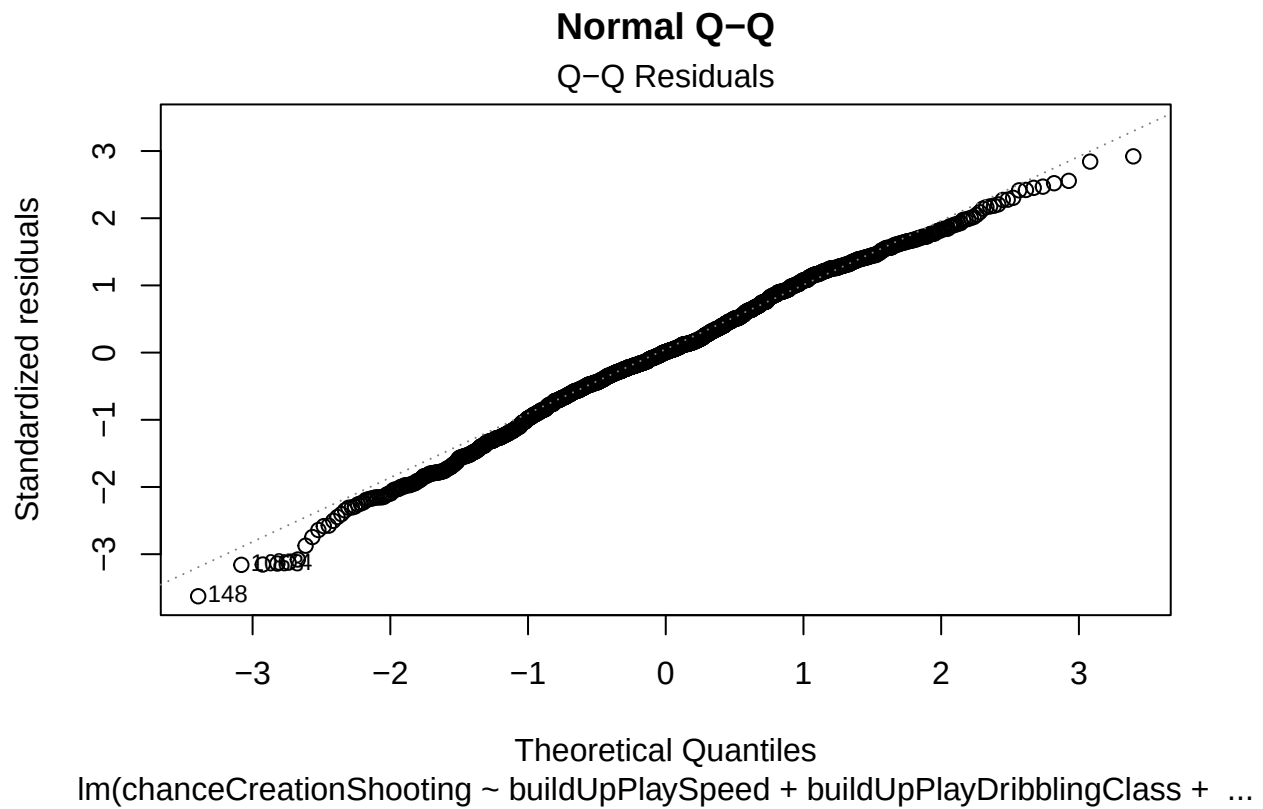
To assess whether the multiple linear regression assumptions are reasonably satisfied, diagnostic plots were examined for linearity, normality of residuals, constant variance, and potentially influential observations.

```
# Model diagnostic plots

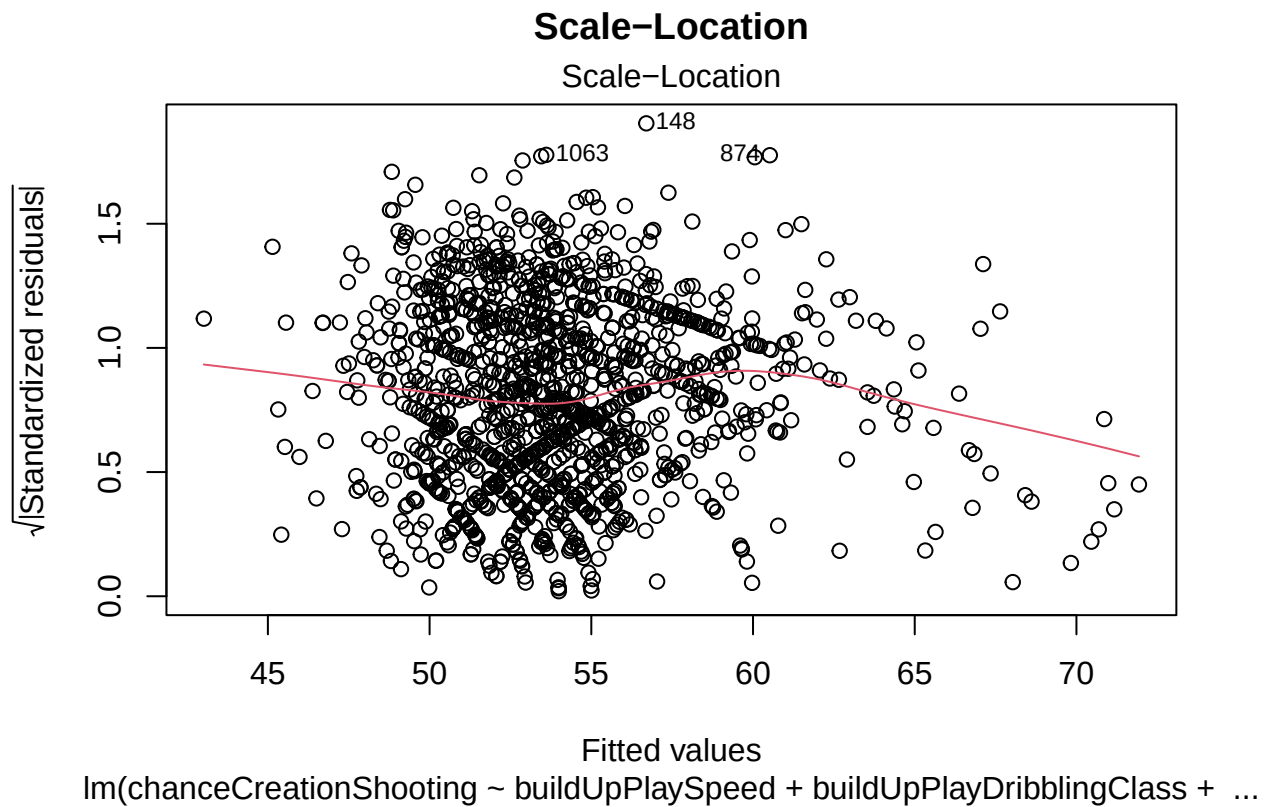
# 1. Residuals vs Fitted
plot(
  soccer_model,
  which = 1,
  main = "Residuals vs Fitted"
)
```



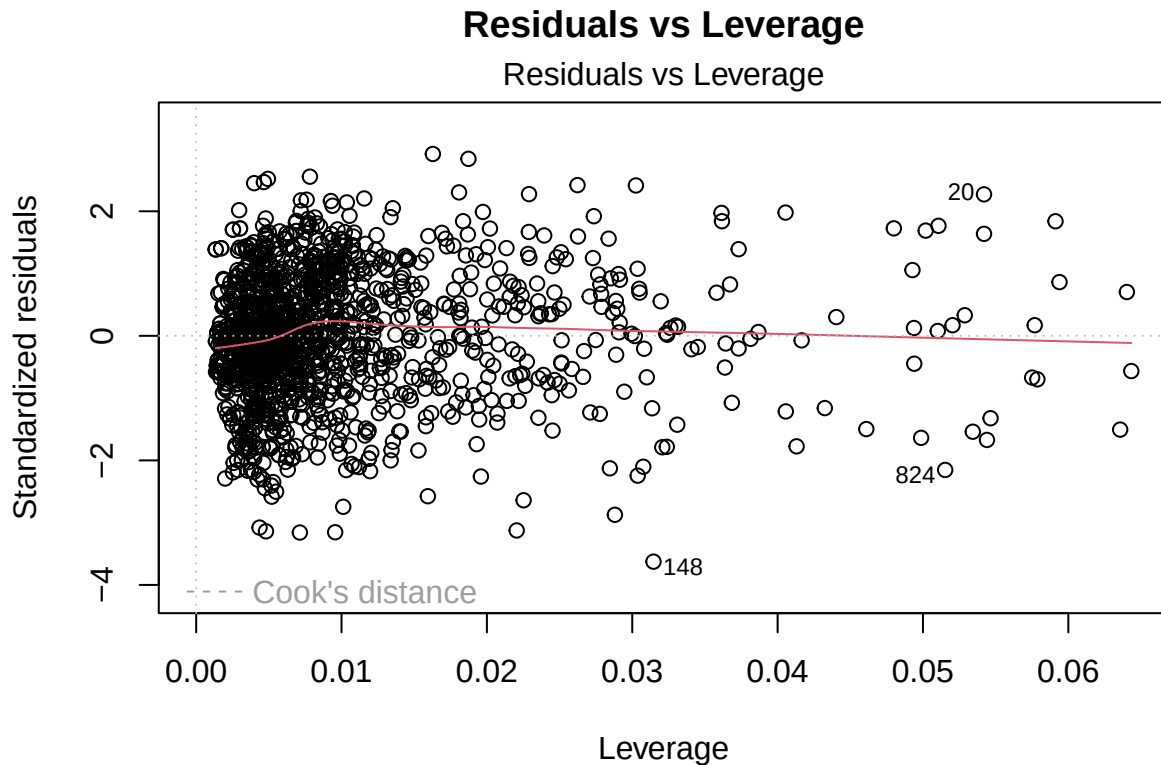
```
# 2. Normal Q-Q  
plot(  
  soccer_model,  
  which = 2,  
  main = "Normal Q-Q"  
)
```



```
# 3. Scale-Location
plot(
  soccer_model,
  which = 3,
  main = "Scale-Location"
)
```



```
# 4. Residuals vs Leverage
plot(
  soccer_model,
  which = 5,
  main = "Residuals vs Leverage"
)
```



`lm(chanceCreationShooting ~ buildUpPlaySpeed + buildUpPlayDribblingClass + ...`

Influential Observation Analysis

Potentially influential observations were evaluated using Cook's distance and leverage measures.

```
# Calculate Cook's distance
cooks_d <- cooks.distance(soccer_model)

# Number of observations
n <- nobs(soccer_model)

# Common Cook's distance threshold
cook_threshold <- 4 / n

# Identify observations exceeding the threshold
influential_obs <- which(cooks_d > cook_threshold)

# Report threshold and number of potentially influential observations
cook_threshold
```

```
## [1] 0.002743484
```

```
length(influential_obs)
```

```
## [1] 92
```

```
# Display the 10 observations with the largest Cook's distances
head(sort(cooks_d, decreasing = TRUE), 10)
```

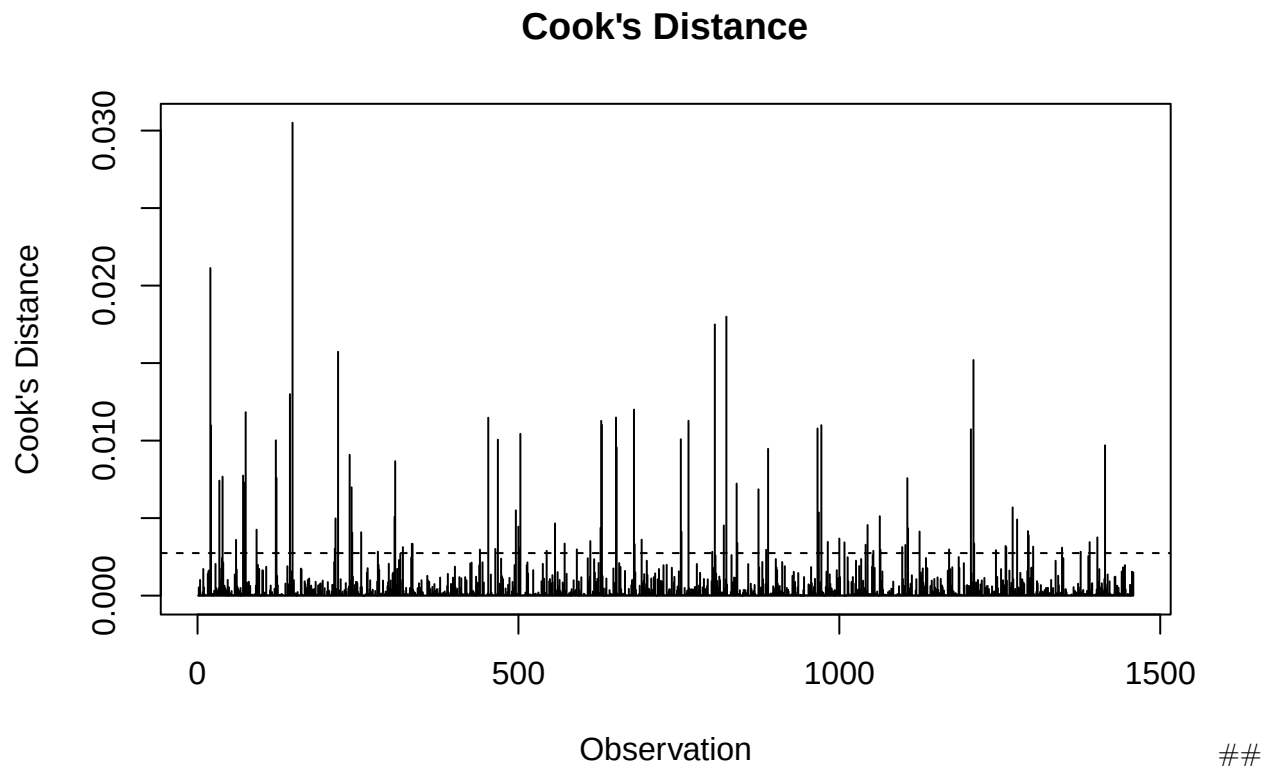
```
##      148      20      824      806      219     1209     144
## 0.03050657 0.02113138 0.01799759 0.01749450 0.01572998 0.01519655 0.01299559
##      680      75      652
## 0.01200260 0.01183147 0.01148673
```

```

# Plot Cook's distance
plot(
  cooks_d,
  type = "h",
  main = "Cook's Distance",
  xlab = "Observation",
  ylab = "Cook's Distance"
)

abline(
  h = cook_threshold,
  lty = 2
)

```



Interaction Effect Analysis

The interaction between chance-creation passing and positioning style was examined to determine whether the association between passing and shooting differs between Free Form and Organised teams.

```

# Extract model coefficients
b <- coef(soccer_model)

# Estimated passing slope for Free Form teams
slope_freeform <- b["chanceCreationPassing"]

# Estimated passing slope for Organised teams
slope_organised <-
  b["chanceCreationPassing"] +
  b["chanceCreationPassing:chanceCreationPositioningClassOrganised"]

# Difference in passing slopes between positioning styles

```

```

slope_difference <-
  b["chanceCreationPassing:chanceCreationPositioningClassOrganised"]

# Display estimated slopes
slope_freeform

## chanceCreationPassing
##          0.3505298
slope_organised

## chanceCreationPassing
##          0.03314887
slope_difference

## chanceCreationPassing:chanceCreationPositioningClassOrganised
##                                -0.3173809

# 95% confidence intervals for the Free Form slope
# and the interaction coefficient
confint(soccer_model)[
  c(
    "chanceCreationPassing",
    "chanceCreationPassing:chanceCreationPositioningClassOrganised"
  ),
]

##                                2.5 %
## chanceCreationPassing          0.2221953
## chanceCreationPassing:chanceCreationPositioningClassOrganised -0.4544682
##                                97.5 %
## chanceCreationPassing          0.4788642
## chanceCreationPassing:chanceCreationPositioningClassOrganised -0.1802936

# Statistical inference for the passing slope among Organised teams

V <- vcov(soccer_model)

b_pass <- coef(soccer_model)["chanceCreationPassing"]
b_int <- coef(soccer_model)[
  "chanceCreationPassing:chanceCreationPositioningClassOrganised"
]

# Estimated passing slope for Organised teams
slope_organised <- b_pass + b_int

# Standard error of the combined slope
se_organised <- sqrt(
  V["chanceCreationPassing", "chanceCreationPassing"] +
  V["chanceCreationPassing:chanceCreationPositioningClassOrganised",
    "chanceCreationPassing:chanceCreationPositioningClassOrganised"] +
  2 * V["chanceCreationPassing",
    "chanceCreationPassing:chanceCreationPositioningClassOrganised"]
)

# t statistic

```

```

t_organised <- slope_organised / se_organised

# p-value
p_organised <- 2 * pt(
  abs(t_organised),
  df = df.residual(soccer_model),
  lower.tail = FALSE
)

# 95% confidence interval
ci_organised <- slope_organised +
  c(-1, 1) *
  qt(0.975, df.residual(soccer_model)) *
  se_organised

# Display results
slope_organised

## chanceCreationPassing
##           0.03314887
se_organised

## [1] 0.02911885
t_organised

## chanceCreationPassing
##           1.138399
p_organised

## chanceCreationPassing
##           0.2551428
ci_organised

## [1] -0.02397091  0.09026864

# Sensitivity analysis excluding observations flagged by Cook's Distance

# Create a sensitivity-analysis dataset
soccer_model_data_sensitivity <- soccer_model_data[-influential_obs, ]

# Refit the same model
soccer_model_sensitivity <- lm(
  chanceCreationShooting ~
    buildUpPlaySpeed +
    buildUpPlayDribblingClass +
    buildUpPlayPassing +
    buildUpPlayPositioningClass +
    chanceCreationPassing * chanceCreationPositioningClass +
    chanceCreationCrossing +
    defencePressure +
    defenceAggression +
    defenceTeamWidth +
    defenceDefenderLineClass,

```

```

data = soccer_model_data_sensitivity
)

# Check remaining sample size
nobs(soccer_model_sensitivity)

## [1] 1366

# Display sensitivity-model results
summary(soccer_model_sensitivity)

##
## Call:
## lm(formula = chanceCreationShooting ~ buildUpPlaySpeed + buildUpPlayDribblingClass +
##     buildUpPlayPassing + buildUpPlayPositioningClass + chanceCreationPassing *
##     chanceCreationPositioningClass + chanceCreationCrossing +
##     defencePressure + defenceAggression + defenceTeamWidth +
##     defenceDefenderLineClass, data = soccer_model_data_sensitivity)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -25.4023  -5.4561  -0.0669   6.1474  24.6240
##
## Coefficients:
##                                     Estimate
## (Intercept)                        23.65199
## buildUpPlaySpeed                     0.10236
## buildUpPlayDribblingClassLots        -3.31317
## buildUpPlayDribblingClassNormal      -2.99254
## buildUpPlayPassing                  -0.06799
## buildUpPlayPositioningClassOrganised -4.22524
## chanceCreationPassing                0.39963
## chanceCreationPositioningClassOrganised 21.29868
## chanceCreationCrossing              -0.05633
## defencePressure                     0.14433
## defenceAggression                   0.06411
## defenceTeamWidth                    0.09030
## defenceDefenderLineClassOffside Trap  5.12259
## chanceCreationPassing:chanceCreationPositioningClassOrganised -0.39566
##                                     Std. Error
## (Intercept)                        4.70650
## buildUpPlaySpeed                    0.02374
## buildUpPlayDribblingClassLots       2.93890
## buildUpPlayDribblingClassNormal     0.52493
## buildUpPlayPassing                  0.02602
## buildUpPlayPositioningClassOrganised 1.33825
## chanceCreationPassing               0.07743
## chanceCreationPositioningClassOrganised 4.45417
## chanceCreationCrossing              0.02311
## defencePressure                     0.03042
## defenceAggression                   0.02807
## defenceTeamWidth                    0.02942
## defenceDefenderLineClassOffside Trap 1.04875
## chanceCreationPassing:chanceCreationPositioningClassOrganised 0.08090
##                                     t value Pr(>|t|)

```

```

## (Intercept) 5.025 5.70e-07
## buildUpPlaySpeed 4.312 1.73e-05
## buildUpPlayDribblingClassLots -1.127 0.25979
## buildUpPlayDribblingClassNormal -5.701 1.46e-08
## buildUpPlayPassing -2.613 0.00907
## buildUpPlayPositioningClassOrganised -3.157 0.00163
## chanceCreationPassing 5.161 2.82e-07
## chanceCreationPositioningClassOrganised 4.782 1.93e-06
## chanceCreationCrossing -2.438 0.01492
## defencePressure 4.745 2.31e-06
## defenceAggression 2.284 0.02253
## defenceTeamWidth 3.069 0.00219
## defenceDefenderLineClassOffside Trap 4.884 1.16e-06
## chanceCreationPassing:chanceCreationPositioningClassOrganised -4.891 1.13e-06
##
## (Intercept) ***
## buildUpPlaySpeed ***
## buildUpPlayDribblingClassLots ***
## buildUpPlayDribblingClassNormal ***
## buildUpPlayPassing **
## buildUpPlayPositioningClassOrganised **
## chanceCreationPassing ***
## chanceCreationPositioningClassOrganised ***
## chanceCreationCrossing *
## defencePressure ***
## defenceAggression *
## defenceTeamWidth **
## defenceDefenderLineClassOffside Trap ***
## chanceCreationPassing:chanceCreationPositioningClassOrganised ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.694 on 1352 degrees of freedom
## Multiple R-squared:  0.1687, Adjusted R-squared:  0.1607
## F-statistic: 21.1 on 13 and 1352 DF, p-value: < 2.2e-16
# Organised-team passing slope in the sensitivity model

V_sens <- vcov(soccer_model_sensitivity)

b_pass_sens <- coef(soccer_model_sensitivity)["chanceCreationPassing"]

b_int_sens <- coef(soccer_model_sensitivity)[
  "chanceCreationPassing:chanceCreationPositioningClassOrganised"
]

# Estimated passing slope for Organised teams
slope_organised_sens <- b_pass_sens + b_int_sens

# Standard error
se_organised_sens <- sqrt(
  V_sens["chanceCreationPassing", "chanceCreationPassing"] +
  V_sens[
    "chanceCreationPassing:chanceCreationPositioningClassOrganised",

```

```

    "chanceCreationPassing:chanceCreationPositioningClassOrganised"
  ] +
  2 * V_sens[
    "chanceCreationPassing",
    "chanceCreationPassing:chanceCreationPositioningClassOrganised"
  ]
)

# t statistic and p-value
t_organised_sens <- slope_organised_sens / se_organised_sens

p_organised_sens <- 2 * pt(
  abs(t_organised_sens),
  df = df.residual(soccer_model_sensitivity),
  lower.tail = FALSE
)

# 95% confidence interval
ci_organised_sens <- slope_organised_sens +
  c(-1, 1) *
  qt(0.975, df.residual(soccer_model_sensitivity)) *
  se_organised_sens

# Display results
slope_organised_sens

## chanceCreationPassing
##          0.003969081
se_organised_sens

## [1] 0.02711079
t_organised_sens

## chanceCreationPassing
##          0.1464022
p_organised_sens

## chanceCreationPassing
##          0.8836257
ci_organised_sens

## [1] -0.04921471  0.05715287

```